

PREDICTIVE ANALYTICS USING HISTORICAL DATA

Objective

The objective of this project is to forecast future sales using historical sales data. Predictive analytics helps organizations anticipate future trends and make informed business decisions.

Tools Used

- Python
- Pandas
- Scikit-Learn
- Matplotlib
- Power BI

Methodology

Historical monthly sales data was collected and preprocessed. A Linear Regression model was trained using historical sales records. The model was then used to forecast sales for the next six months. Model performance was evaluated using the R^2 score.

Key Findings

1. Historical sales exhibited a consistent upward trend throughout the year.
2. The Linear Regression model successfully captured the trend and generated future sales predictions.
3. Forecasted sales indicate continued business growth over the next six months.
4. The predictive model provides useful insights for inventory planning, budgeting, and strategic decision-making.
5. Visualization of actual and predicted sales helps stakeholders understand future business performance.

Conclusion

Predictive analytics transforms historical data into actionable forecasts. The project demonstrates how machine learning can support business planning and improve decision-making by predicting future trends accurately.